

```
In [46]: import pyodbc
import pandas as pd

server = r'DESKTOP-PEBE499\SQLEXPRESS'
database = 'finale_project'
conn_str = f'DRIVER={{SQL Server}};SERVER={server};DATABASE={database};Trusted_Connection=yes;'
conn = pyodbc.connect(conn_str)

print("Connected Successfully! ✅")
```

Connected Successfully! ✅

```
In [47]: import warnings
warnings.filterwarnings('ignore')
```

SQL finale project

Q1:

Write a query to display gender, age, education level, and internet type for all students.

```
In [48]: query = """select gender, age, education_level,internet_type from
students s inner join learning_environment le
ON s.student_id = le.student_id"""
df = pd.read_sql(query, conn)
df
```

```
Out[48]:
```

	gender	age	education_level	internet_type
0	Boy	21-25	University	Wifi
1	Girl	21-25	University	Mobile Data
2	Girl	16-20	College	Wifi
3	Girl	11-15	School	Mobile Data
4	Girl	16-20	School	Mobile Data
...	...	...	...	...
495	Girl	16-20	College	Wifi
496	Boy	11-15	School	Mobile Data
497	Boy	21-25	University	Wifi
498	Boy	21-25	University	Wifi
499	Boy	21-25	University	Wifi

500 rows × 4 columns

Q2:

Display all students who use Wifi and have Moderate adaptivity level. Show gender, education level, and device.

```
In [49]: query = """select gender, education_level, device from students s
inner join learning_environment le ON s.student_id = le.student_id
where internet_type = 'Wifi' and adaptivity_level = 'Moderate'"""
df = pd.read_sql(query, conn)
df
```

```
Out[49]:
```

	gender	education_level	device
0	Boy	University	Tab
1	Girl	College	Mobile
2	Boy	School	Mobile
3	Girl	College	Mobile
4	Boy	College	Mobile
...	...	...	...
96	Girl	University	Computer
97	Girl	School	Mobile
98	Boy	University	Mobile
99	Boy	University	Computer
100	Boy	University	Computer

101 rows × 3 columns

Q3:

Show all school students who use Mobile Data and have Low adaptivity level. Display gender, age, and network type

```
In [50]: query = """select gender, age, network_type from students s
inner join learning_environment le ON s.student_id = le.student_id
where internet_type = 'Mobile Data' and adaptivity_level = 'Low'"""
df = pd.read_sql(query, conn)
df
```

Out[50]:

	gender	age	network_type
0	Girl	16-20	3.0
1	Boy	11-15	3.0
2	Girl	26-30	4.0
3	Boy	6-10	4.0
4	Girl	16-20	3.0
...	...	...	...
120	Girl	11-15	3.0
121	Boy	16-20	4.0
122	Boy	16-20	3.0
123	Boy	21-25	4.0
124	Boy	11-15	3.0

125 rows × 3 columns

Q4:

Display the number of students for each internet type and network type.

```
In [51]: query = """select count (s.student_id) ,le.internet_type,le.network_type
from students s inner join learning_environment le
ON s.student_id = le.student_id group by network_type,internet_type"""
df = pd.read_sql(query, conn)
df
```

Out[51]:

		internet_type	network_type
0	6	Mobile Data	2.0
1	142	Mobile Data	3.0
2	144	Mobile Data	4.0
3	1	Wifi	2.0
4	34	Wifi	3.0
5	173	Wifi	4.0

Q5:

Show education level, device, and number of students. Include only students who have Poor OR Mid financial condition.

```
In [52]: query = """ select education_level, device ,count (s.student_id) as
number_of_students from students s inner join learning_environment le ON s.studer
where financial_condition = 'poor' or financial_condition = 'Mid' group by educati
df = pd.read_sql(query, conn)
```

df

Out[52]:

	education_level	device	number_of_students
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0	College	Computer	2
1	School	Computer	8
2	University	Computer	49
3	College	Mobile	81
4	School	Mobile	196
5	University	Mobile	130
6	College	Tab	1
7	University	Tab	7

Q6:

Display education level and the average class duration for each education level. Only show education levels where the average class duration is greater than 1 hour

```
In [53]: query = """select s.education_level , avg(cast (left(le.class_duration,1)as float )
as Average_Duration from  students s inner join learning_environment le
ON s.student_id = le.student_id group by s.education_level
HAVING AVG(CAST(LEFT(le.Class_Duration, 1) AS FLOAT)) > 1"""

df = pd.read_sql(query, conn)

df
```

Out[53]:

	education_level	Average_Duration
--	-----------------	------------------

0	College	1.211765
1	University	1.580000

Q7:

Display device and the count of students. Only include devices that are used by more than one student.

```
In [54]: query = """ select count (s.student_id ) as number_of_students , le.device
from students s inner join learning_environment le ON s.student_id = le.student_id
group by device HAVING COUNT(s.student_id) > 1"""

df = pd.read_sql(query, conn)

df
```

Out[54]:

	number_of_students	device
0	68	Computer
1	423	Mobile
2	9	Tab

Q8:

Write a query using CASE to create a new column called Internet_Quality: • If Internet Type is Wifi and Network Type is 4G, show Good • Otherwise, show Limited Display gender, education level, internet type, network type, and Internet_Quality.

In [55]:

```
query = """select s.gender, s.education_level, le.internet_type, le.network_type,
case when le.Internet_Type = 'Wifi' and le.Network_Type = '4.00' then 'Good'
else 'Limited' end as Internet_Quality from students s inner join learning_env
ON s.student_id = le.student_id"""

df = pd.read_sql(query, conn)

df
```

Out[55]:

	gender	education_level	internet_type	network_type	Internet_Quality
0	Boy	University	Wifi	4.0	Good
1	Girl	University	Mobile Data	4.0	Limited
2	Girl	College	Wifi	4.0	Good
3	Girl	School	Mobile Data	4.0	Limited
4	Girl	School	Mobile Data	3.0	Limited
...	...	...	...	...	...
495	Girl	College	Wifi	4.0	Good
496	Boy	School	Mobile Data	3.0	Limited
497	Boy	University	Wifi	3.0	Limited
498	Boy	University	Wifi	4.0	Good
499	Boy	University	Wifi	4.0	Good

500 rows × 5 columns

Q9:

Create a VIEW called connected_students that includes students who: • Have Location = Yes • AND use Wifi Then display all records from the view ordered by Adaptivity Level

In [56]:

```
query = """ select s.student_id ,le.Adaptivity_Level from students s
inner join learning_environment le ON s.student_id = le.student_id
where le.location =1 and internet_type ='Wifi' SELECT * FROM connected_students
ORDER BY
```

```

CASE Adaptivity_Level
WHEN 'High' THEN 1
WHEN 'Moderate' THEN 2
WHEN 'Low' THEN 3
ELSE 4 END ASC"""

df = pd.read_sql(query, conn)

df

```

Out[56]:

	student_id	Adaptivity_Level
0	100	Moderate
1	102	Moderate
2	106	Low
3	107	Moderate
4	108	Low
...	...	...
186	589	Moderate
187	590	High
188	593	High
189	597	Moderate
190	598	Moderate

191 rows × 2 columns

Q10:

For each education level, display: • Total number of students • Number of students with Low adaptivity level Only show education levels where more than 30% of students have Low adaptivity level

```

In [57]: query = """select education_level, total_students,
low_count from( SELECT  s.education_level,
COUNT(s.student_id) AS total_students,
SUM(CASE WHEN le.adaptivity_level = 'Low' THEN 1
ELSE 0 END) AS low_count FROM students s INNER JOIN
learning_environment le ON s.student_id = le.student_id
GROUP BY s.education_level)as subquery
WHERE (CAST(low_count AS FLOAT) / total_students) > 0.30"""

df = pd.read_sql(query, conn)

df

```

Out[57]:

	education_level	total_students	low_count
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0	College	85	46
1	School	215	82
2	University	200	80